ANALYZE: Exploratory Data Analysis

The goal of the EDA session is to aggregate data on groups, identify counts and averages, and order data to determine trends amount different groups, i.e. times, type of riders, etc. The final produce resulting from this session a firm understanding of trends that can fuel marketing efforts of rental e-bikes.

Query Log:

- Query 25: Verify row count
- Query 26: Preview processed features
- Query 27: Compare total rides by user type
- Query 28: Compare rides by day of week and user type
- Query 29: Compare rides by hour of day and user type
- Query 30: Calculate average ride length by user type
- Query 31: Count rides over 30 minutes by user type
- Query 32: Compare rides by trip type and user type
- Query 33: Compare rides by bike type and user type
- Query 34: Rank top 10 start stations for casuals
- Query 35: Rank top 10 end stations for casuals

Query 25: Verify row count

```
None
SQL
SELECT COUNT(*) AS total_rows
FROM `cyclistic-case-study-493121.CycleData.q1_2025_processed`
```

Result/Observation: validating that all data was aggregated appropriately and no corruption occurred.

Query 26: Preview processed features

```
None
SQL
SELECT
        ride_id,
        member_casual,
        ride_length_minutes,
        day_number,
        hour_number,
        trip_type
FROM `cyclistic-case-study-493121.CycleData.q1_2025_processed`
```

```
LIMIT 10;
```

Result/Observation: ensuring that all engineered data is properly loaded and is as meant to be.

Query 27: Compare total rides by user type

```
None
SQL
SELECT
        member_casual,
        COUNT(*) AS total_rides
FROM `cyclistic-case-study-493121.CycleData.q1_2025_processed`
GROUP BY member_casual;
```

Result/Observation: able to compare how many total rides are in each member and casual riders.

Query 28: Compare rides by day of week and user type

```
None
SQL
SELECT
        member_casual,
        day_number,
        COUNT(*) AS total_rides
FROM `cyclistic-case-study-493121.CycleData.q1_2025_processed`
GROUP BY member_casual, day_number
ORDER BY member_casual, day_number;
```

Result/Observation:

Identifying how members and casual riders differ based on the day the ride took place.

Query 29: Compare rides by hour of day and user type

```
None
SQL
```

```
SELECT
	member_casual,
	hour_number,
	COUNT(*) AS total_rides
FROM `cyclistic-case-study-493121.CycleData.q1_2025_processed`
GROUP BY member_casual, hour_number
ORDER BY member_casual, hour_number;
```

Result/Observation:

Identifying how member and casual riders compare as to the hour of the day.

Query 30: Calculate average ride length by user type

```
None
SQL
SELECT
	member_casual,
	ROUND(AVG(ride_length_minutes), 2) AS avg_ride_length
FROM `cyclistic-case-study-493121.CycleData.q1_2025_processed`
GROUP BY member_casual;
```

Result/Observation:

Identifying the average ride length in minutes of member riders vs. casual riders.

Query 31: Count rides over 30 minutes by user type

```
None
SQL
SELECT
	member_casual,
	COUNT(*) AS long_rides
FROM `cyclistic-case-study-493121.CycleData.q1_2025_processed`
WHERE ride_length_minutes >= 30
GROUP BY member_casual;
```

Result/Observation:

Identified how many riders in each group, members and casual riders, spend more than 30 rides riding per day.

Query 32: Compare rides by trip type and user type

```
None
SQL
SELECT
        member_casual,
        trip_type,
        COUNT(*) AS total_rides
FROM `cyclistic-case-study-493121.CycleData.q1_2025_processed`
GROUP BY member_casual, trip_type
ORDER BY member_casual, trip_type;
```

Result/Observation:

Identify the number of riders in distinct members and casuals by each trip type, to identify patterns between trip types and whether the riders are members or casual.

Query 33: Compare rides by bike type and user type

```
None
SQL
SELECT
        member_casual,
        rideable_type,
        COUNT(*) AS total_rides
FROM `cyclistic-case-study-493121.CycleData.q1_2025_processed`
GROUP BY member_casual, rideable_type
ORDER BY member_casual, rideable_type;
```

Result/Observation:

Identifying the number of riders in both members and casual riders, distinguished based on rideable type status.

Query 34: Rank top 10 start stations for casuals

```
None
SQL
SELECT start_station_name
```

```
FROM `cyclistic-case-study-493121.CycleData.q1_2025_processed`
WHERE member_casual = 'casual'
GROUP BY start_station_name
ORDER BY start_station_name
LIMIT 10;
```

Result/Observation:

Identfied which which start station has the most casual riders begin their ride with them.

Query 35: Rank top 10 end stations for casuals

```
None
SQL
SELECT end_station_name
FROM `cyclistic-case-study-493121.CycleData.q1_2025_processed`
WHERE member_casual = 'casual'
GROUP BY end_station_name
ORDER BY COUNT(*) DESC
LIMIT 10;
```

Result/Observation: Identify which end station has the most casual riders who end their ride with them.

End of Query Repository